Appendix E

Geometric Algebra and the N-Dimensional Wage

The 2D complex number $W = \psi_m + j\psi_s$ models a single imaginary axis of oppression (e.g., race). Oppression is fractal in nature and operates across multiple simultaneous, orthogonal axes; the framework must therefore be extended to higher-dimensional algebraic structures.

This appendix is a speculative extension, in the same register as the photon model (Appendix F). The algebra supplies a precise formal encoding of the directional claim that intersectional oppression compounds non-additively; it carries no calibrated measurement content, and Tier 3 status applies throughout. The empirical authority of the framework rests on the historical chapters and the tier-tagged case studies, with this material offered as a formal language for future development.

E.1 The Quaternion Stage: Three Intersectional Axes

For three primary imaginary axes (race, gender, sexuality), the standard complex number $a + bj$ is extended to a *quaternion*:

$$W_Q = a + bi + cj + dk \tag{E.1}$$

where i, j, k are three distinct imaginary units satisfying:

$$i^2 = j^2 = k^2 = ijk = -1 \tag{E.2}$$

Component	EE Axis	Sociological Meaning
a	Real axis	Material Wage ψ_m — wealth, land, healthcare
bi	Imaginary axis 1	Racial Status Wage (e.g., whiteness)
cj	Imaginary axis 2	Gendered Status Wage (e.g., patriarchy/maleness)
dk	Imaginary axis 3	Cisnormative/Sexuality Status Wage

Theorem E.1 (Multiplicative Intersection Compounding (Misogynoir))**.** *Because* i, j, k

are mathematically orthogonal and multiply according to quaternion rules:

$$i \times j = k, \quad j \times i = -k \tag{E.3}$$

Under this algebra, intersectional oppression compounds multiplicatively. For a Black woman, the intersection of the racial wage axis (i) and the gendered wage axis (j) multiplies to create a distinct third vector k—a unique plane of systemic extraction requiring both axes for its full specification. This formalizes Kimberlé Crenshaw's theory of intersectionality as a quaternion multiplication law.[1]

E.2 The N-Dimensional Weighted Summation Equation

Beyond three axes, the framework uses *Geometric Algebra* (Clifford Algebra) over N orthogonal imaginary bases $e_1, e_2, \ldots, e_N$. The full Complex Wage is:

$$W = \psi_m + \sum_{k=1}^{N} j_k \left(\rho_k \cdot \psi_{s,k}\right) \tag{E.4}$$

where:

- ψ_m: The Real Material Wage.
- j_k: The k-th imaginary basis axis (e.g., j_1 = race, j_2 = gender, j_3 = ability, j_4 = sexuality, j_5 = neurology, j_6 = height/physicality, ...).
- $\psi_{s,k}$: The baseline psychological wage on axis k.
- ρ_k: The **Demographic Weight Coefficient**—the proportion of the population for whom axis k is a primary axis of marginalization, scaled by institutional bandwidth devoted to that axis.

The Transphobia Paradox (High Voltage, Low Current)

The electrical analog of reactive power is $P = VI$ where V is voltage and I is current (population size). If ρ_k is small (small population directly experiencing a given axis), but the Elite needs large reactive power P_k from that axis to sustain the

[1] This theorem is classified as Tier 3 (ordinal/structural); the quaternion formalism provides a mathematical encoding of the directional claim that intersectional oppression is qualitatively distinct from the sum of its constituent axes. See the Empirical Methodology chapter (p. lxv).

overall pacification of the Buffer Class, then:

$$P_k = V_k \cdot I_k \implies V_k = \frac{P_k}{\rho_k} \tag{E.5}$$

As $\rho_k \to 0$, the required voltage $V_k \to \infty$. This mathematically predicts and explains the disproportionate legislative, media, and political bandwidth (hyper-inflated voltage) devoted to a statistically small demographic node (e.g., trans individuals). The Elite pumps astronomical voltage into a small-current node to generate sufficient reactive power to distract the Buffer Class from macro-level material extraction.[a]

[a]This derivation is classified as Tier 3 (ordinal/structural); it formalizes the directional claim that the Elite can achieve high reactive power from small demographic targets by intensifying the ideological voltage directed at them. See the Empirical Methodology chapter (p. lxv).

E.3 Bivectors and Intersectional Space

In Geometric Algebra, multiplying two basis vectors produces a *bivector*—a distinct two-dimensional plane:

$$e_1 \wedge e_4 = e_{14} \tag{E.6}$$

If e_1 is Race and e_4 is Disability, the bivector e_{14} describes the unique extraction plane of a Black disabled person. This individual is located in e_{14}—a qualitatively distinct geometric zone where extraction is accelerated, enforcement violence is compounded, and systemic neglect is multiplicatively intensified. The bivector e_{14} encodes an extraction plane qualitatively different from either axis alone.

Theorem E.2 (Fractal Consistency of Geometric Algebra)**.** *The Geometric Algebra extension reproduces the same extraction geometry at the multi-axis level that the complex number $W = \psi_m + j\psi_s$ produces at the single-axis level. Specifically:*

- *At the single-axis level, Orthogonal Deflection is produced by the magnetic cross-product $q(\vec{v} \times \vec{B})$: the Buffer Class is deflected horizontally by the psychological wage field.*
- *At the multi-axis level, the same deflection is produced by the superposition of N cultural magnetic fields $\sum_{k=1}^{N} \rho_k \vec{B}_k$, each weighted by its demographic coefficient ρ_k.*

The fractal is exact: the mathematics is the same mathematics operating at different dimensional resolutions.[2]

[2]This theorem is classified as Tier 3 (ordinal/structural). See the Empirical Methodology chapter

(p. lxv).